Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

EPPH808T – MOLECULAR PHYSICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. What do you mean by electronic spectroscopy?
2. State Frank Condon principle.
3. What is Raman Effect?
4. Mention any two applications of Raman spectroscopy.
5. What are prolate and oblate symmetric tops?
6. Give the principle of FTIR spectroscopy.
7. What is the nuclear g_N fact ^{19}F nucleus which has a magnetic moment of 2.6273 μ_N. Nuclear spin quantum number I = ½.
8. Define chemical shift.
9. What is hyperfine structure?
10. State the principle of Mossbauer spectroscopy.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Discuss diatomic molecules as harmonic and anharmonic oscillators.

(or)

b) Explain Fortrat parabola.

12. a) Explain quantum theory of Raman Effect.

(or)

b) Describe Laser Raman spectroscopy.

13. a) Obtain an expression for the vibrational energy of a diatomic molecule.

(or)

b) With necessary diagram explain the instrumentation techniques of a FTIR spectroscopy.

14. a) Discuss the relaxation process of NMR spectroscopy.

(or)

b) Discuss in detail the concept of NMR spectroscopy.

15. a) Explain the function of a Mossbauer spectrometer.

(or)

b) State the principle of nuclear quadrupole resonance.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Describe the rotational fine structure of electronic-vibration spectra.

17. Explain vibration-rotation Raman spectra of a diatomic molecule.

18. Describe the theory of diatomic vibrating rotator and draw the rotational energy level diagram.

19. Draw a diagram for NMR instrumentation and explain.

20. With a block diagram explain ESR spectrometer.

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